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The Loyola Index Family and the Resolution Limit of Degree Imbalance

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Abstract

For a graph G, the Gutman–Milovanović index $M_{\alpha,\beta}(G)$ sums $(d_u d_v)^\alpha (d_u + d_v)^\beta$ over the edges uv and unifies the Zagreb and Randić indices. The *Loyola index family* $LO(G;\alpha,\beta,\gamma)$ multiplies the term of each edge by $e^{\gamma q_{uv}}$, where $q_{uv} = |d_u - d_v|/(d_u + d_v)$ is its normalised degree imbalance, so that $LO(G;\alpha,\beta,0) = M_{\alpha,\beta}(G)$. By theorems of Rada, the points $LO(G;0,0,\gamma)$ at algebraic $\gamma \neq 0$ separate exactly the histograms of edge imbalances, and no degree-ratio index, such as the degree-ratio Sombor index, separates more. By exhaustive enumeration, this histogram determines the degree-pair counts of a tree up to order twelve (fifteen for chemical trees), with explicit counterexamples at every larger order. We also show that two chemical graphs exchange order at most five times along γ, and that the path and the star are the extremal trees of the pure-γ points. Pre-registered tests on measured boiling points did not demonstrate a predictive gain from γ. A gain on vaporisation enthalpies that met a pre-registered seed-count rule shrinks to about zero, in an exploratory comparison, once the parent is searched over a box that does not truncate it.

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1 Introduction

Degree-based topological indices form one of the most developed descriptor classes of chemical graph theory. The class runs from the Randić and Zagreb indices [25,43] and their parametric extensions, the general Randić and sum-connectivity families [8,56], to the Sombor index [19] and its many variants, surveyed in [20]. Extremal results on bond-incident-degree (BID) indices are surveyed in [2], and a finite-dimensional linear framework for such indices is developed in [40].

The *Gutman–Milovanović family* [22], studied in the recent bounds and QSPR literature [18, 33, 38], is $M_{\alpha,\beta}(G) = \sum_{uv\in E(G)}(d_u d_v)^{\alpha}(d_u + d_v)^{\beta}$, $\alpha, \beta \in \mathbb{R}$. Its special cases, catalogued in [33], include $M_{0,1} = M_1$, $M_{1,0} = M_2$, the hyper-Zagreb index $M_{0,2} = HM$ [48], $M_{-1,0} = {}^{m}M_2$, the Randić index $M_{-1/2,0} = R$, the sum-connectivity index $M_{0,-1/2} = \chi$ [55], the harmonic index $2M_{0,-1} = H$, the inverse sum indeg index $M_{1,-1} = ISI$, the geometric–arithmetic index $2M_{1/2,-1} = GA$ [51] and the arithmetic–geometric index $\frac{1}{2}M_{-1/2,1} = AG$. Its two parameter axes are the general Randić and sum-connectivity families, and parameter-optimised versions have already been fitted to octane properties [3].

We study the three-parameter extension

$$LO(G;\alpha,\beta,\gamma) = \sum_{uv\in E(G)} (d_u d_v)^{\alpha}(d_u + d_v)^{\beta} e^{\gamma q_{uv}}, \qquad q_{uv} = \frac{|d_u - d_v|}{d_u + d_v},$$

the *Loyola index family*, with $LO(G;\alpha,\beta,0) = M_{\alpha,\beta}(G)$. Positive γ upweights degree-imbalanced edges, negative γ downweights them, and on regular graphs γ has no effect. We ask what γ adds. It reads no graph information beyond the degree-pair counts, but at rational parent exponents it can separate graphs that the parent merges.

The family is a three-parameter subfamily of the exponential vertex-degree-based indices of Rada [39], whose extremal theory is developed in [11, 17, 49] and, for general edge weights, in [26]. Published exponential indices [13, 14] exponentiate a single classical generator. The variable Euler–Sombor index [24] and the cosine-rule Sombor indices [36] weight the imbalance polynomially and without normalisation. Functions of the

ratio q_{uv} appear in the diminished Sombor index [34], in degree-ratio indices [21,37], and, as a normalised form of Albertson's irregularity [1,30], in $IRLA(G) = 2\sum_{uv} q_{uv}$ [45]. The pure-γ point $LO(G;0,0,\gamma) = \sum_{uv} e^{\gamma q_{uv}}$ is the exponential index, in the sense of [39], of the edge function γq_{uv}. The coordinate added here is linear in q_{uv} in the exponent and is coupled with its own parameter to the Gutman–Milovanović family. It makes every edge weight log-linear in (α, β, γ), so LO is the partition function of an exponential family on the edges of G.

On graphs with $\Delta \leq 4$ every BID index is a linear functional of the ten edge-degree-pair counts, a representation used in [39, 41] and developed systematically in [40], so the whole family lies in a fixed ten-dimensional space. Our questions concern a nonlinear family inside that space: which count vectors a parameter point separates, and how the index moves along γ. The tools are classical. Gutman's survey [20] observes that new variants hardly ever improve on earlier ones in applications, and our tests agree. The family is not proposed as a better predictor, and the QSPR sets used here are small.

Several facts used below are known. Graphs with equal counts receive equal values of every vertex-degree-based index, and attaching the same pendant path to two such graphs gives examples of every larger order [40, Section 3.1]. An index whose edge weights coincide on a class of degree pairs depends on a graph only through the class sums of its counts [39, Proposition 2.5], and the exponential of an index with algebraic edge weights separates exactly these class sums [39, Theorem 3.4]. The path and the star are the extremal trees of the pure-γ points by the criteria of Cruz and Rada [12]. What is new here is where these facts bind for chemical graphs. The q-histogram carries the full degree-pair information of an alkane carbon skeleton up to order 15 and not beyond, so no degree-ratio index, such as GA, AG or the diminished Sombor index, separates all skeletons of order 16 or more. At rational (α, β) a nonzero algebraic γ never merges two molecules that the parent index separates, and the order of two chemical graphs changes at most five times as γ varies.

Our contributions are as follows.

(i) Two elementary facts: the imbalance direction is affinely indepen-

dent of the parent coordinates once a vertex of degree at least 3 is admitted (Lemma 1), and $\log LO$ is a log-sum-exp and hence convex (Theorem 1).

(ii) The ratio plane Π (Theorem 2), on which the family depends on a graph only through its q-histogram. That the pure-γ points at algebraic $\gamma \neq 0$ separate exactly the q-histograms is Theorem 3.4 of [39] applied to the weights γq_{ij}, and that no degree-ratio index separates more is Proposition 2.5 there. New are explicit trees with distinct profiles and equal q-histograms at every order from 13, and from 16 for $\Delta \leq 4$, and the computer-assisted result that these orders are the first. The corollary carries the collisions to every degree-ratio index, such as GA, AG and the diminished Sombor index.

(iii) A bound on the number of order exchanges along γ (Proposition 4) and per-graph identifiability (Proposition 5).

(iv) Two octane collision pairs realising the known degree-pair ceiling [40, Section 3.1] (Proposition 7), value counts for trees of orders 7 to 17, and structure sensitivity (Section 4).

(v) Closed forms on standard graphs, the extremal trees of the pure-γ points, which follow from the criteria of Cruz and Rada [12], and termwise bounds (Section 5).

(vi) A matched-budget QSPR diagnostic and pre-registered tests on measured boiling points (Section 6), in which freeing γ gives no demonstrated gain once the parent's search box does not truncate it (Section 6.2). On measured boiling points ridge regression on the degree-pair counts leads every single index evaluated in point estimate (Section 6.4).

2 Preliminaries and notation

Let $G = (V, E)$ be a finite simple connected graph with at least one edge, of order $n = |V|$ and size $m = |E|$, and for a vertex $v \in V$ let the degree $d_v = d_G(v)$ count the edges incident to v. The maximum and minimum

degrees are denoted $\Delta = \Delta(G)$ and $\delta = \delta(G)$. The chemometric sections use hydrogen-suppressed molecular graphs of alkane carbon skeletons, for which $\Delta \le 4$, while the mathematical results do not assume this bound unless stated.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair*, understood as an unordered multiset $\{d_u, d_v\}$. For integers $1 \le i \le j$ let $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ be the number of edges with degree pair (i, j), called (i, j)-edges; the vector $(m_{ij}(G))_{i \le j}$ is the *degree-pair profile* (or count vector) of G.

Definition 1 (Loyola index family)**.** Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as

$$LO(G; \alpha, \beta, \gamma) \;=\; \sum_{uv \in E(G)} (d_u d_v)^{\alpha} \, (d_u + d_v)^{\beta} \, \exp\!\left(\gamma \, \frac{|d_u - d_v|}{d_u + d_v} \right) \tag{1}$$

Eq. (1) is defined for all $\alpha, \beta, \gamma \in \mathbb{R}$ on graphs without isolated vertices. Write $q_{uv} = \mathrm{imb}(d_u, d_v)$, where $\mathrm{imb}(x, y) = |x - y|/(x + y)$, for the normalised degree imbalance of the edge uv, and $q_{ij} = (j - i)/(i + j)$ for $i \le j$. Each summand depends only on the unordered degree pair of its edge, so

$$LO(G; \alpha, \beta, \gamma) = \sum_{i \le j} m_{ij}(G) \, (ij)^{\alpha} (i + j)^{\beta} e^{\gamma q_{ij}} . \tag{2}$$

In particular $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$, the Gutman–Milovanović family [22, 33], and $LO(G; \alpha, \beta, \gamma) = M_{\alpha,\beta}(G)$ for every γ when G is regular.

Definition 2 (q-histogram)**.** The q-histogram of G is the function $n_q(G) = \sum_{i \le j,\; q_{ij} = q} m_{ij}(G)$, $q \in [0, 1)$, the number of edges of G with imbalance q.

Equal profiles give equal q-histograms. The converse fails (Theorem 2(iii)).

3 Parameter geometry

Throughout this section $\theta = (\alpha, \beta, \gamma)$ and $z_{uv} = \big(\ln(d_u d_v), \ln(d_u + d_v), q_{uv}\big)$, so that

$$LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}. \tag{3}$$

For a degree pair (i, j) we write z_{ij}, and call $(1, z_{ij}) \in \mathbb{R}^4$ its augmented feature vector. A set of augmented vectors spans $\mathbb{R}^4$ if and only if the corresponding z_{ij} do not lie in one affine plane of $\mathbb{R}^3$.

Lemma 1 (Affine independence of the imbalance direction)**.** *For $\Delta \geq 3$, with $P_\Delta = \{(i,j) : 1 \leq i \leq j \leq \Delta\}$ the admissible degree pairs, the augmented feature vectors $\big(1, \ln(ij), \ln(i+j), q_{ij}\big)$, $(i,j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples θ, θ' satisfy $e^{\theta^\top z} = c\, e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i,j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$. In particular, no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, while for $\Delta = 1$ the only pair is $(1,1)$, with augmented vector $(1, 0, \ln 2, 0)$, and the span has dimension 1. In both cases the conclusion fails. For $\Delta = 2$, indeed, $q_{ij} = [\ln(i+j) - \ln 2 - \frac{1}{2}\ln(ij)]/(3\ln 3 - \frac{9}{2}\ln 2)$ on the three pairs, so for $\Delta = 2$ the factor $e^{\gamma q_{ij}}$ equals a constant times $(ij)^a (i+j)^b$ on the three pairs, for suitable constants a, b. The effect of γ is therefore reproduced exactly by a shift of (α, β) together with a scaling constant.*

Proof. The vectors $(1, \ln(ij), \ln(i+j))$ on the pairs $(1,1), (2,2), (1,2)$ have rank 3, since the corresponding 3×3 determinant equals $\ln 2 \cdot \ln(9/8) \neq 0$, so it suffices to show that q_{ij} is not an affine function of $\big(\ln(ij), \ln(i+j)\big)$ on P_Δ. The four pairs $(1,1), (2,2), (1,2), (1,3)$, all admissible once $\Delta \geq 3$, witness this. The pair $(1,1)$ occurs only in K_2, and the conclusion persists without it, since the augmented vectors of $(1,2), (1,3), (2,2), (3,3)$, namely $(1, \ln 2, \ln 3, \frac{1}{3})$, $(1, \ln 3, \ln 4, \frac{1}{2})$, $(1, \ln 4, \ln 4, 0)$ and $(1, \ln 9, \ln 6, 0)$, have determinant $-\frac{1}{6}\ln\frac{3}{2}\,\ln\big(2^{13}/3^8\big) \approx -0.0150 \neq 0$. Suppose $q_{ij} = a + b\ln(ij) + c\ln(i+j)$ on these pairs. The pairs $(1,1)$ and $(2,2)$ have $q = 0$, so they give $a + c\ln 2 = 0$ and $a + (2b + 2c)\ln 2 = 0$, hence $a = -c\ln 2$ and $b = -c/2$. Substituting into the $(1,2)$ and $(1,3)$ equations gives $\frac{1}{3} = c\,(\ln 3 - \frac{3}{2}\ln 2)$ and $\frac{1}{2} = c\,(\ln 2 - \frac{1}{2}\ln 3)$. In particular $c \neq 0$, since the

left side $\frac{1}{3}$ is nonzero. Three times the first minus twice the second gives $0 = c\,(4\ln 3 - \frac{13}{2}\ln 2)$, hence $8\ln 3 = 13\ln 2$, i.e. $3^8 = 2^{13}$. This is false ($6561 \neq 8192$). Hence the augmented vectors have rank 4. Indeed, a linear relation among the four coordinates on P_Δ with a nonzero coefficient on q_{ij} would express q_{ij} as an affine function of $\ln(ij)$ and $\ln(i{+}j)$, which we have just excluded. A relation with zero coefficient on q_{ij} is excluded by the rank-3 determinant. If $e^{\theta^\top z} = c\,e^{\theta'^\top z}$ on P_Δ, then $(\theta-\theta')^\top z_{ij} = \ln c$ for all admissible pairs, so $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector. Since the augmented vectors span $\mathbb{R}^4$, the only vector orthogonal to all of them is 0, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1,1), (1,2), (2,2)$ exist. The 3×3 determinant computed at the start of the proof shows that their three augmented vectors have rank 3, and three vectors cannot have more. For $\Delta = 1$ the single augmented vector has rank 1. ∎

Remark (Independence versus per-graph identifiability). Lemma 1 concerns the edge-weight *function* on the full admissible pair set. On a single graph G the index can merely rescale along a direction v, $LO(G;\theta+tv) = e^{ct}\,LO(G;\theta)$, exactly when the feature vectors z_{ij} of its edges fail to affinely span $\mathbb{R}^3$ (Proposition 5). On an r-regular graph, for instance, $LO(G;\theta) = m\,e^{\theta^\top z_0}$ with $z_0 = (2\ln r, \ln 2r, 0)$, so the values determine only $\theta^\top z_0$. The lemma guarantees that no such degeneracy is forced once all pairs with $\Delta \geq 3$ are available.

By (3), $\log LO_G$ is a log-sum-exp function of θ.

Theorem 1 (Log-convexity)**.** *Fix a graph G with $m \geq 1$ edges, define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$, and let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}}/LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \qquad \nabla^2 F_G(\theta) = \mathrm{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0,1)$ and $\partial_\gamma^2 F_G = \mathrm{Var}_\theta(q) \geq 0$, so $\log LO$ is non-decreasing and convex in γ, and it is strictly increasing in γ unless every edge of G joins vertices of equal degree. Moreover $\mathrm{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G,

which covers regular graphs ($q \equiv 0$) and, for example, (Δ, δ)-semiregular bipartite graphs with $\Delta > \delta$. A bipartite graph is (Δ, δ)-semiregular when every vertex on one side has degree Δ and every vertex on the other side has degree δ, so every edge has degree pair (δ, Δ). On these every edge carries the same nonzero imbalance, and $\log LO$ is affine and strictly increasing in γ. F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv \in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$. On a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\mathrm{Cov}_\theta(z) = 0$.

Proof. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k]\,\mathbb{E}_\theta[z_l]$, the (k,l)-entry of $\mathrm{Cov}_\theta(z)$, which is positive semidefinite. Convexity follows. The measure $w(\theta)$ has full support on E for every θ. Hence $\mathrm{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace. If they do lie in such a subspace, then along the corresponding v the function $t \mapsto F_G(\theta + tv)$ is affine, so F_G is not strictly convex. If they do not, the Hessian is positive definite at every θ, which implies strict convexity. This proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r-regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ∎

The Cauchy–Schwarz interpolation corollary of Section 5.3 is the midpoint case of the convexity of F_G.

3.1 The ratio plane, crossings and identifiability

Since $4ij/(i+j)^2 = 1 - q_{ij}^2$, every admissible pair satisfies $\ln(ij) = 2\ln(i+j) + \ln(1 - q_{ij}^2) - \ln 4$, and therefore

$$LO(G; \alpha, \beta, \gamma) = 4^{-\alpha} \sum_{uv \in E(G)} (d_u + d_v)^{2\alpha+\beta}\,(1 - q_{uv}^2)^\alpha\, e^{\gamma q_{uv}}. \qquad (4)$$

With $\kappa = 2\alpha + \beta$, the parameter κ acts only on the degree sum $d_u + d_v$, and α, γ act only on q_{uv}, a function of the degree ratio d_u/d_v.

Definition 3 (Ratio plane)**.** The ratio plane is $\Pi = \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : \beta = -2\alpha\}$, the plane $\kappa = 0$. It contains the pure-γ points $(0, 0, \gamma)$, $GA =$

$2\,LO(\frac{1}{2},-1,0)$ and $AG=\frac{1}{2}LO(-\frac{1}{2},1,0)$.

Graphs with equal profiles receive equal values of every vertex-degree-based index [39,40], and sufficient linear criteria for an index to separate all distinct profiles within a class of graphs of fixed order are given in [40,41]. The theorem below concerns a coarser loss. On Π distinct profiles with equal q-histograms receive equal values. Part (iii) shows that such profiles occur among trees of every order from 13 (from 16 when $\Delta \le 4$); that no smaller tree has them is a computer-assisted result (see the remark on the floor on Π later in this subsection).

Theorem 2 (Ratio plane)**.** *Let G and H be graphs.*

(i) On Π, $LO(G;\alpha,-2\alpha,\gamma)=\sum_q n_q(G)\left((1-q^2)/4\right)^{\alpha}e^{\gamma q}$ depends on G only through its q-histogram, and G and H agree at every point of Π if and only if $n_q(G)=n_q(H)$ for all q. Otherwise their values differ at every point of Π outside a set of measure zero.

(ii) For every nonzero algebraic γ, $LO(G;0,0,\gamma)=LO(H;0,0,\gamma)$ if and only if $n_q(G)=n_q(H)$ for all q. On any finite set of graphs the same holds for all but finitely many real γ.

(iii) Let G and H be trees with distinct edge-degree-pair profiles and equal q-histograms, each having a leaf adjacent to a vertex of the same degree $d\ge 2$. Attaching a pendant path of length $L\ge 1$ at such a leaf in each produces trees with the same two properties and the same maximum degree. The profiles change by the same vector, whether or not the chosen leaf edge was the only $(1,d)$-edge. Consequently, for every $n\ge 13$ (and every $n\ge 16$ among trees with $\Delta\le 4$) there are non-isomorphic trees of order n with distinct degree-pair profiles that no point of Π separates.

Proof. (i) The formula is immediate from (4), since setting $\beta=-2\alpha$ removes the degree-sum factor, and it shows that equal histograms give agreement at every point of Π. For the measure-zero clause, suppose the histograms differ. At $\alpha=0$ the difference is $\sum_q(n_q(G)-n_q(H))\,e^{\gamma q}$, an exponential sum with distinct exponents and not all coefficients zero, so it is not identically zero. This uses the first of two standard facts: a

nonzero exponential sum with distinct real exponents is not identically zero, because as $\gamma \to +\infty$ the term with the largest exponent dominates. The difference on Π is therefore a nonzero real-analytic function of (α, γ), whose zero set has measure zero. This is the second fact: the zero set of a nonzero real-analytic function on $\mathbb{R}^2$ has Lebesgue measure zero [32]. (ii) Agreement for equal histograms is immediate from the formula in (i). Since degrees are integers, every imbalance q is rational, so for algebraic $\gamma \neq 0$ the exponents γq, $q \in \mathbb{Q}$, are distinct algebraic numbers. The Lindemann–Weierstrass theorem [5] says that for distinct algebraic numbers $x_1, \dots, x_k$ the exponentials $e^{x_1}, \dots, e^{x_k}$ are linearly independent over the field of algebraic numbers. Hence the numbers $e^{\gamma q}$ are linearly independent over the algebraic numbers, and the integer coefficients $n_q(G) - n_q(H)$ must all vanish. For real γ, a nonzero exponential sum with k distinct real exponents has at most $k - 1$ real zeros (Proposition 4 below), and a finite set of graphs gives finitely many such sums. (iii) The last clause follows from (i) once the trees exist, since equal histograms give agreement on all of Π. The attachment, a device also used by Rada [40] for edge-type vectors, changes the profile of each tree by the same vector, $-(1,d) + (2,d) + (L-1)(2,2) + (1,2)$, so distinct profiles stay distinct and equal histograms stay equal. The new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. Trees realising the two order-13 profiles listed in Section 4.2 are the following. Their edge lists confirm the common q-histogram displayed there: one edge of imbalance 0, two of $\frac{1}{5}$, one of $\frac{1}{3}$, three of $\frac{1}{2}$ and five of $\frac{5}{7}$. T_1: a path $c\,a\,b\,d\,e\,\ell$ in which c carries five pendant leaves, b and d carry one pendant leaf each and ℓ is a leaf, so that $d_c = 6$, $d_a = d_e = 2$, $d_b = d_d = 3$; its edges are $(1,6)^5, (2,6), (2,3)^2, (3,3), (1,3)^2, (1,2)$. T_2: a path $c\,b\,x\,y\,z$ in which c carries five pendant leaves, b one and z two, so that $d_c = 6$, $d_b = d_z = 3$, $d_x = d_y = 2$; its edges are $(1,6)^5, (3,6), (1,3)^3, (2,3)^2, (2,2)$. Both have a leaf adjacent to a vertex of degree $d = 3$. Consider now the order-16 pair with $\Delta \leq 4$. The profile $\{(1,2)^2, (1,4)^8, (2,4)^2, (4,4)^3\}$ is realised by a path $A\,B\,C\,D$ of vertices of degree 4. Here A and D carry three pendant leaves each, while B and C carry one pendant leaf and one pendant path of length 2 each. The profile $\{(1,4)^8, (2,2)^3, (2,4)^4\}$ is realised by three

vertices A, B, C of degree 4. They are joined as $A\,w\,B$ through a vertex w of degree 2 and as $B\,v_1v_2v_3v_4\,C$ through a path of four vertices of degree 2. The vertices A, B and C carry three, two and three pendant leaves. Both trees have the q-histogram $\{0^3, \frac{1}{3}^4, \frac{3}{5}^8\}$, since the pairs $(1,2)$ and $(2,4)$ have $q = \frac{1}{3}$, the pair $(1,4)$ has $q = \frac{3}{5}$ and the pairs $(2,2)$ and $(4,4)$ have $q = 0$. Both also have a leaf adjacent to a vertex of degree $d = 4$, which gives every larger order. ∎

Part (ii) is Theorem 3.4 of [39] applied to the algebraic edge weights γq_{ij}, whose classes of equal weight are the q-classes: two graphs receive the same value for every algebraic $\gamma \neq 0$ exactly when they have equal q-histograms. Part (i) follows directly from the factorisation (4). The statement for real γ applies the standard fact that a nonzero exponential sum has finitely many real zeros. The propagation by a pendant path in part (iii) is the device of [40]. The construction parallels Example 2.2 of [41], where indices that depend only on the degree sum are defeated by chemical trees whose counts agree on classes of equal sum. Here the classes are those of equal ratio, which for $\Delta \leq 4$ are $\{(1,2),(2,4)\}$ and $\{(2,2),(3,3),(4,4)\}$ (ignoring $(1,1)$, which occurs only in K_2). A class-wide question of the same kind was studied by Song and Huang [50] for distance-type invariants. The edge division vector of a tree T counts its edges e by the order $\min\{n_u(e), n_v(e)\}$ of the smaller component of $T - e$. Trees with equal vectors receive equal values of every edge-additive eccentric index $\sum_e f(\min\{n_u(e), n_v(e)\})$, such as the Wiener and Gutman indices, and Song and Huang determine all such trees of order at most 10. That invariant does not control degrees: the two seven-vertex trees with edge division vector $(4,1,1)$ have different degree-pair profiles and different q-histograms, while the two octane pairs of Table 1 have different edge division vectors and different Wiener indices. Neither partition of trees refines the other, so their collisions and ours are of different kinds. Explicit trees of this kind for a single index were given by Rada [39, Example 2.6], who exhibits two trees on 29 vertices that have different class counts and equal Randić index. Rada, Rodríguez and Sigarreta [41, Example 2.1] give an infinite family of chemical trees with equal Randić index and different counts. We found no earlier determination of the orders 13 and 16, or

of explicit colliding trees for degree-ratio indices, in a literature search recorded in the repository [35]. The minimality of orders 13 and 16 is computer-assisted. It rests on an exhaustive enumeration of all trees of orders 2 to 16 in exact rational arithmetic, reproducible with the code of the repository [35].

Corollary (Degree-ratio indices)**.** *Let* $\rho_{uv} = \min(d_u, d_v)/\max(d_u, d_v)$ *and let* $I_f(G) = \sum_{uv \in E(G)} f(\rho_{uv})$ *for any function* f *on* $(0, 1]$*. Since* $\rho_{uv} = (1-q_{uv})/(1+q_{uv})$*,* I_f *depends on* G *only through its* q*-histogram. Consequently the trees of Theorem 2(iii), which exist at every order* $n \geq 13$ *(and* $n \geq 16$ *with* $\Delta \leq 4$*), are not separated by any such index. This applies to GA, AG, the exponential geometric–arithmetic index [13], the diminished Sombor index [34], the symmetric division deg index [52], the inverse symmetric division deg index and its exponential [14], and the degree-ratio indices [21, 37], including the degree-ratio Sombor index studied in [4, 23, 47].*

Remark. The corollary follows from the identity for ρ_{uv}; it is the structure observation of [39, Proposition 2.5] for edge weights that depend only on the degree ratio, and what is new are the explicit witnesses and the orders at which they first occur. Within the class it describes, the pure-γ points at algebraic $\gamma \neq 0$ attain the most possible: by part (ii) they separate every pair of graphs with distinct q-histograms. GA does not (see the remark on the floor on Π below). The parent attains it as well: at $\gamma = 0$ part (i) gives $\sum_q n_q\,((1-q^2)/4)^\alpha$, an exponential sum in α with distinct bases, which separates two distinct histograms at all but finitely many α. The limit thus belongs to the dependence on the degree ratio, not to γ, which adds the form valid at every algebraic value. The orders 13 and 16 are the first at which the q-histogram, and hence every degree-ratio index simultaneously, fails to separate distinct profiles. A single index can fail earlier through coincidences among its values. The symmetric division deg index merges two distinct profiles of chemical trees already at order 9, for instance $\{(1,2),(1,3)^2,(2,2)^4,(2,3)\}$ and $\{(1,2)^3,(1,3),(2,3)^3,(3,3)\}$, both with value 58/3, and the inverse symmetric division deg index does so at order 10 (exact enumeration in the repository).

Off Π the same transcendence argument gives the following.

Proposition 3 (Parent distinctions survive)**.** *Let $\alpha, \beta \in \mathbb{Q}$ and let $\gamma \neq 0$ be algebraic. For graphs G, H and each imbalance value q write $c_q(G,H) = \sum_{(i,j):\, q_{ij}=q} \big(m_{ij}(G) - m_{ij}(H)\big)(ij)^\alpha (i+j)^\beta$. Then $LO(G;\alpha,\beta,\gamma) = LO(H;\alpha,\beta,\gamma)$ if and only if $c_q(G,H) = 0$ for every q. In particular, if $M_{\alpha,\beta}(G) \neq M_{\alpha,\beta}(H)$ then $LO(G;\alpha,\beta,\gamma) \neq LO(H;\alpha,\beta,\gamma)$: at rational exponents a nonzero algebraic γ never destroys a distinction made by the parent.*

Proof. The difference is $\sum_q c_q(G,H)\, e^{\gamma q}$, a sum over the distinct imbalances q occurring in G or H. For rational α, β the numbers $(ij)^\alpha (i+j)^\beta$ are algebraic, being rational powers of positive integers, so the coefficients $c_q(G,H)$, integer combinations of them, are algebraic. For algebraic $\gamma \neq 0$ the exponents γq are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [5] the sum vanishes only if every $c_q(G,H)$ vanishes. The last claim follows because $\sum_q c_q(G,H) = M_{\alpha,\beta}(G) - M_{\alpha,\beta}(H)$. ∎

Remark. At rational α, β, two graphs merged by $M_{\alpha,\beta}$ stay merged at an algebraic $\gamma \neq 0$ only if every class sum $c_q(G,H)$ vanishes, which is stronger than $M_{\alpha,\beta}(G) = M_{\alpha,\beta}(H)$. On the trees of order 16, $\gamma = 1$ recovers the distinctions the parent loses at $(0, \frac{1}{2}, 0)$ and $(0, \frac{1}{3}, 0)$ (Section 4.2). The algebraic hypothesis cannot be dropped: at $\gamma = \ln 4$, $LO(K_4; 0, 0, \gamma) = 6 = 3e^{\gamma/2} = LO(K_{1,3}; 0, 0, \gamma)$.

Remark (The floor on Π). By exhaustive enumeration of all trees of orders 2–17 in exact arithmetic (Table 2), two trees of order at most 12 with equal q-histograms have equal degree-pair profiles, and the same holds for trees with $\Delta \leq 4$ of order at most 15. With part (iii), the pure-γ points at algebraic $\gamma \neq 0$ attain the degree-pair floor of Section 4.1, separation of all distinct profiles, exactly up to those orders and at no larger order. Among all trees, at orders 13–17, generic points off Π (Proposition 8) and the Randić index take more distinct values on these trees than any point of Π. A point of Π can also do worse. GA merges two distinct q-histograms among trees first at order 17, for instance $\{(1,5)^{10}, (2,2)^2, (5,5)^2, (2,5), (1,2)\}$ and $\{(4,5)^3, (1,4)^5, (1,5)^6, (2,5), (1,2)\}$, which have equal GA and different q-histograms.

Proposition 4 (Order exchanges along γ)**.** *Fix (α,β) and graphs G, H, let $q_1<\cdots<q_r$ be the distinct imbalances of the degree pairs occurring in G or H, and write $D(\gamma)=LO(G;\alpha,\beta,\gamma)-LO(H;\alpha,\beta,\gamma)=\sum_{k=1}^{r}c_k e^{\gamma q_k}$ with $c_k=\sum_{q_{ij}=q_k}(m_{ij}(G)-m_{ij}(H))(ij)^{\alpha}(i+j)^{\beta}$. Let V be the number of sign changes in $(c_1,\dots,c_r)$ after zero terms are removed. If $D\not\equiv 0$, then D has at most $V\le r-1$ real zeros counted with multiplicity, and their number has the parity of V. In particular, for graphs with $\Delta\le 4$ the possible imbalances are $0,\frac{1}{7},\frac{1}{5},\frac{1}{3},\frac{1}{2},\frac{3}{5}$, and two such graphs exchange order at most five times as γ ranges over $\mathbb{R}$, for every (α,β).*

Proof. With $x=e^{\gamma}>0$, D becomes the generalised polynomial $\sum_k c_k x^{q_k}$ with distinct real exponents, so Laguerre's extension of Descartes' rule of signs applies [27]. Consider a finite sum $\sum_k c_k x^{q_k}$ with real exponents $q_1<\cdots<q_r$ and let V count the sign changes in its coefficient sequence. The rule says that the number of zeros in $x>0$, counted with multiplicity, is at most V and differs from V by an even number. Zeros of D in γ correspond one-to-one, with multiplicity, to positive zeros in x. The number of real zeros of D, with multiplicity, is therefore at most V and has its parity. As an independent check, the parity also follows from the signs of D as $\gamma\to\mp\infty$, which are the signs of the first and last nonzero c_k. Divide D by $e^{\gamma q_k}$ for the first k with $c_k\neq 0$ and let $\gamma\to-\infty$, or for the last such k and let $\gamma\to+\infty$. As in the proof of Proposition 6 below, only c_k survives. ■

Remark (Sharpness)*.* The bound is attained. The trees of order 12 with profiles $\{(1,3)^2,(1,4)^2,(1,6)^5,(3,4),(4,6)\}$ and $\{(1,2),(1,5)^4,(1,6)^4,(2,6),(5,6)\}$ have $\Delta=6$, eight distinct imbalances and $V=5$, and at $(\alpha,\beta)=(-1,0)$ they cross at $\gamma\approx-17.2$, -12.0, -2.0, 2.7 and 31.1, which are therefore all zeros of D. The first tree is a vertex of degree 6 with five leaves and a neighbour of degree 4, which carries two leaves and a vertex of degree 3 with two leaves. The second is a vertex of degree 6 with four leaves, a neighbour of degree 5 with four leaves, and a neighbour of degree 2 ending in a leaf. Whether five crossings are attained for $\Delta\le 4$ is open. Along a line $\theta+tv$ not parallel to the γ axis the exponents of D are the values $v^{\top}z_{ij}$, of which there can be as many as admissible pairs, so the constant five is specific to the γ axis.

For a graph G let $R(G)$ be the set of degree pairs it realises, $r(G)$ the number of distinct imbalances among them, and $A(G)$ the affine dimension of $\{z_{ij} : (i,j) \in R(G)\}$, that is, the dimension of the smallest affine subspace of $\mathbb{R}^3$ containing it.

Proposition 5 (Per-graph identifiability)**.** *(i) $A(G)$ equals the rank of the Hessian $\mathrm{Cov}_\theta(z)$ of $\log LO_G$ at every θ, and the directions v for which there is a constant $c = c(v)$ with $LO(G;\theta+tv) = e^{ct}\,LO(G;\theta)$ for all t and θ form a subspace of dimension $3 - A(G)$.*

(ii) $A(G) \le \min\big(|R(G)| - 1,\ r(G),\ 3\big)$ for every graph.

(iii) If $r(G) = 2$ with imbalances $q_1 \neq q_2$, then $v = (1, -2, c)$ with $c = [\ln(1-q_2^2) - \ln(1-q_1^2)]/(q_1 - q_2)$ is such a direction.

Proof. (i) The measure $w(\theta)$ puts positive weight on every edge, so, as in the proof of Theorem 1, $\mathrm{Cov}_\theta(z)$ is singular exactly in the directions v with $v^\top z$ constant on $R(G)$. Such v are those orthogonal to the differences of the z_{ij}, and they form a subspace of dimension $3 - A(G)$, so the rank of $\mathrm{Cov}_\theta(z)$ is $A(G)$. (In the language of exponential families [9], $w(\theta)$ is the family with sufficient statistic z on the support $R(G)$. The rank statement says that its minimal representation has $A(G)$ parameters.) A direction v rescales the index, $LO(G;\theta+tv) = e^{ct}LO(G;\theta)$ for all θ and t, exactly when $v^\top \nabla F_G(\theta) = c$ is independent of θ. Differentiating, this holds exactly when $\mathrm{Cov}_\theta(z)\, v = 0$ for all θ, that is, when $v^\top z$ is constant on $R(G)$. The rescaling directions are therefore the subspace of dimension $3 - A(G)$ just described. (ii) By (4), $z_{ij} = (2s + \ln(1-q^2) - \ln 4,\ s,\ q)$ with $s = \ln(i+j)$, so pairs of equal imbalance lie on one line parallel to $(2,1,0)$, and r parallel lines span an affine space of dimension at most r. (iii) $v^\top z_{ij} = \ln(1-q_{ij}^2) - \ln 4 + c\,q_{ij}$ takes the same value at q_1 and q_2 by the choice of c. ∎

Remark (The case $\Delta = 7$). Equality holds in (ii) for every set of admissible pairs with $\Delta \le 6$ (by exhaustive enumeration), hence for all molecular graphs. It fails from $\Delta = 7$: among the subsets of P_7 with at most four pairs, the only one violating equality is $\{(1,7),(2,6),(3,5),(4,4)\}$, whose degree sums all equal 8. Every violating set contains a violating subset of

at most four pairs. If $A(R) = a$ is below the bound, then $a \le 2$, and R has at least $a+1$ imbalances and $a+2$ pairs. One pair from each of $a+1$ imbalance classes and one further pair form a subset S of $a+2 \le 4$ pairs, with $A(S) \le a$ while its bound is $a+1$. Such a subset consists of three collinear points with two imbalances or four coplanar points with three imbalances. Any fifth pair of P_7 has a degree sum different from 8 and leaves that plane, so this is the only violating set. A constant degree sum makes $\ln(i+j)$ constant on the set, so $A \le 2$, while the bound equals 3 as soon as the set has at least four pairs and three imbalances. In a connected graph whose edges all have degree sum s, each neighbour of a vertex of degree d has degree $s-d$, so only the pair $\{d, s-d\}$ occurs. Hence no connected graph realises this set.

Example 1. The spider with four legs of length three (a centre vertex joined to four paths on three vertices) realises $(2,4)$, $(2,2)$ and $(1,2)$, with the two imbalances $\frac{1}{3}$ and 0. On it γ is confounded with (α, β) along the direction of part (iii).

Remark (Data sets). For a QSPR data set the relevant set is the union of the pairs realised by its molecules. A direction that rescales every molecule's index by the same factor leaves every correlation unchanged and is absorbed into a regression slope. On the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so the octane fits of Section 6 are free of exact confounding of this kind.

Proposition 6 (Limits along the γ axis)**.** *Let G have at least one edge, let $E_{\max}(G)$ denote the set of edges realising $\mathrm{imb}_{\max}(G) = \max_{uv} \mathrm{imb}(d_u, d_v)$ and $E_{\min}(G)$ the set realising $\mathrm{imb}_{\min}(G) = \min_{uv} \mathrm{imb}(d_u, d_v)$. Then as $\gamma \to +\infty$,*

$$\frac{LO(G;\alpha,\beta,\gamma)}{e^{\gamma\,\mathrm{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^{\alpha} (d_u + d_v)^{\beta},$$

and as $\gamma \to -\infty$,

$$LO(G;\alpha,\beta,\gamma)/e^{\gamma\,\mathrm{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^{\alpha} (d_u + d_v)^{\beta}.$$

Proof. With $\mu(G) := \mathrm{imb}_{\max}(G)$ and $\varepsilon_{uv} := \mu(G) - \mathrm{imb}(d_u, d_v) \geq 0$, we have $LO(G;\alpha,\beta,\gamma)/e^{\gamma\mu(G)} = \sum_{uv}(d_u d_v)^{\alpha}(d_u + d_v)^{\beta}e^{-\gamma\varepsilon_{uv}}$. The summands over $E_{\max}(G)$ ($\varepsilon_{uv} = 0$) are constant, and the finitely many others tend to 0 as $\gamma \to +\infty$. The case $\gamma \to -\infty$ is dual. ■

4 Discrimination

4.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal degree-pair counts, so the number of distinct profiles is a *ceiling* on distinct values and a *floor* on degeneracy.

Proposition 7 (Degree-pair ceiling)**.** *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$ for all $i \leq j$, then $LO(G;\theta) = LO(H;\theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI. Among the* 18 *octane isomers exactly* 16 *distinct count vectors occur, and the pairs* {*3-methylheptane, 4-methylheptane*} *and* {*3,4-dimethylhexane, 3-ethyl-2-methylpentane*} *share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible in-sample error on this benchmark.*

Proof. By Definition 1, $LO(G;\theta)$ depends on G only through the multiset of unordered edge degree pairs, that is, through $(m_{ij}(G))_{i\leq j}$, and the same holds for every vertex-degree-based index [40], since these graphs share the M-polynomial [15]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package, and Table 1 lists the common count vectors of both collision pairs. ■

A function of the count vector predicts one value per collision class, so its sum of squared errors is at least the within-class sum of squares. For the octane boiling points this floor is $2.925\,^{\circ}\mathrm{C}^2$, a root mean square of $0.403\,^{\circ}\mathrm{C}$. It bounds a single fixed predictor, not the cross-validated errors of Section 6, where each fold fits its own.

Remark (Dimension of the counts). For a tree of order $n \geq 3$ with $\Delta \leq 4$ the ten counts are not free. A $(1,1)$ edge would be a component K_2, so $m_{11} = 0$. There are $n-1$ edges, and since a vertex of degree d lies on d edges, $\sum_{i \leq j} (1/i + 1/j)\, m_{ij} = n$ (see [12, eqs. (3)–(4)]). The nine remaining counts therefore have affine dimension at most 7 at fixed n and at most 8 when n varies. Both bounds are attained by the data sets of Section 6: the centred count matrices of the 18 octanes and the 34 measured decanes have rank 7, and that of the 100 pooled alkanes has rank 8.

The ceiling is tight. Distinct count vectors are separated at almost every parameter choice.

Proposition 8 (Generic discrimination). *Fix $\Delta \geq 1$ and let $m \neq m'$ be two distinct edge-degree-pair count vectors supported on the admissible pairs P_Δ. Then $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij})\, e^{\theta^\top z_{ij}}$ is a real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$ [32]. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot\,;\theta)$ for almost every $\theta \in \mathbb{R}^3$, while only graphs with identical count vectors (Proposition 7) are indistinguishable for all θ.*

Proof. The map $(i,j) \mapsto z_{ij}$ is injective on P_Δ, since $\ln(ij)$ and $\ln(i+j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct, which excludes only finitely many hyperplanes of v. Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$ that are not all zero, and such an exponential sum vanishes identically only if all $c_k = 0$. Hence $D \not\equiv 0$ on the ray, and so on $\mathbb{R}^3$. Since D is a finite sum of exponentials of linear functionals, it is real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ∎

In the terminology of Rada, Rodríguez and Sigarreta [41], an index is weakly discriminating over a class when equal values force equal count vectors; they showed that several classical indices fail this over chemical trees while the Sombor index satisfies it. They also showed that weights linearly

independent over $\mathbb{Z}$, for instance exponentials of distinct algebraic numbers, give weakly discriminating indices [41, Theorems 3.1 and 3.8], and support restrictions of this criterion are given in [40, Theorem 3, Corollary 2]. As there are countably many pairs of count vectors, Proposition 8 gives that almost every point of the family is weakly discriminating over every class at once.

The proof uses only the injectivity of $(i, j) \mapsto (\ln(ij), \ln(i + j))$ and therefore applies to the parent $M_{\alpha,\beta}$ on the (α, β)-plane, so Proposition 8 is inherited from the parent rather than conferred by γ. Since $q_{ij} = \sqrt{1 - 4ij/(i + j)^2}$ depends only on the pair, γ adds no graph information beyond the edge-degree-pair counts, of which every vertex-degree-based index is a functional [40]. What it adds is the functional direction of Lemma 1, a different weighting of the same information.

Proposition 8 concerns generic real exponents. The indices used in practice have rational exponents, and there the parent can fail: the sum-connectivity index $\chi = M_{0,-1/2}$ and the index $M_{0,1/2}$ with edge weight $\sqrt{d_u + d_v}$ are not weakly discriminating over chemical trees [41, Example 2.2]. The imbalance coordinate separates the pair of trees used there for χ.

Example 2. For $t \geq 1$ the chemical trees S and T of order $17 + t$ in Example 2.2 of [41] have $m(S) - m(T) = 6(1, 3) + 4(2, 3) + 4(3, 3) - 4(1, 4) - 6(2, 2) - 4(2, 4)$ and equal sum-connectivity index. At $(\alpha, \beta) = (0, -\frac{1}{2})$,

$$D(\gamma) = LO(S) - LO(T) = 3\big(e^{\gamma/2} - 1\big) - \tfrac{4}{\sqrt{5}}\big(e^{3\gamma/5} - e^{\gamma/5}\big) + \tfrac{4}{\sqrt{6}}\big(1 - e^{\gamma/3}\big),$$

so $D(0) = 0$ and $D(1) \approx 0.2255$. The separation is small for small γ, for instance $D(10^{-6}) \approx 2.4 \times 10^{-7}$.

4.2 Degeneracy across tree orders

Exponential indices of this type were introduced by Rada [39], who studied their discriminating power.

Degeneracy [28] measures the failure to discriminate non-isomorphic graphs, and we quantify it with the distinct-value convention $100(1 - k/N)$,

Table 1. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector, shown as multiplicities of the unordered pairs, and is therefore indistinguishable by every vertex-degree-based index for all parameter values. The two members differ in the measured properties. Units: T_B in °C, ΔH_f and ΔH_{vap} in kcal mol^{-1}, S in cal mol^{-1} K^{-1}; ω is dimensionless.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
shared count vector $2(1,2)$, $1(1,3)$, $2(2,2)$, $2(2,3)$:					
3-methylheptane	118.9	−50.82	9.52	111.26	0.371
4-methylheptane	117.7	−50.69	9.48	109.32	0.372
shared count vector $2(1,2)$, $2(1,3)$, $2(2,3)$, $1(3,3)$:					
3,4-dimethylhexane	117.7	−50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	−50.48	9.21	106.06	0.330

where k is the number of distinct values among $N = 106$ trees (Supplementary Section S5). It is not the share of trees whose value another tree also takes, which is larger: for $LO(0,0,1)$, 45 of the 106 trees (42.5%). On the 106 trees of order 10 the pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$ attain 25.5% (79 distinct values), tied with R, GA and AG. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the floor for *any* degree-pair-based index, while M_1 (83.0%) and M_2 (67.0%) are the most degenerate (Figure 1). For orders 7 through 12, on all trees and on those with $\Delta \leq 4$ (Supplementary Table S4), the pure-γ points and R attain the floor at every order. The index χ leaves it from order 10, with 77 values against 79 profiles. Theorem 2 makes this exact. The pure-γ points separate precisely the q-histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. At order 13 the profiles $\{(1,2)^1,(1,3)^2,(1,6)^5,(2,3)^2,(2,6)^1,(3,3)^1\}$ and $\{(1,3)^3,(1,6)^5,(2,2)^1,(2,3)^2,(3,6)^1\}$ (exponents are edge multiplicities) share the q-histogram $\{0^1,\frac{1}{5}^2,\frac{1}{3}^1,\frac{1}{2}^3,\frac{5}{7}^5\}$. Table 2 collects the first failures, and the index R still attains the floor at order 13 (570 values) and leaves it at order 14, where the pure-γ points give 1090 values.

Off the ratio plane the imbalance factor can restore discrimination lost at particular classical exponents. A point $(0,\beta,0)$ weights an edge by its degree sum only, so pairs with equal sum receive equal weight. With $\gamma \neq 0$

Table 2. First tree order at which discrimination is lost. Values are counted as distinct at 11 significant digits. Recounts at 50 to 60 digits agree. Every count, ties included, is also certified exactly. For the pure-γ points this follows from the q-histograms (Theorem 2(ii)). For R and GA, including the merge of GA at order 17, it follows from exact arithmetic in the radicals $(ij)^a(i+j)^b$. This rests on the linear independence over $\mathbb{Q}$ of the k-th roots of distinct k-th-power-free positive integers [7]. "Below floor" is the first order with fewer distinct values than distinct edge-degree-pair profiles, the floor for every vertex-degree-based index. The entry gives (values/profiles) at that order. "Merges" is the first order at which an index on Π gives equal values to two trees with different q-histograms. This does not apply to R, which is not on Π. $LO(0,0,\gamma)$ denotes the pure-γ points $\gamma = 1, 2$. Orders 2 to 17 were examined. Equal q-histograms of distinct profiles were sought at every order from 2 and first found at the orders shown.

	all trees		trees with $\Delta \leq 4$	
Index	below floor	merges	below floor	merges
$LO(0,0,\gamma)$	13 (566/570)	never[a]	16 (809/810)	never[a]
R	14 (1095/1100)	n/a[b]	17 (1191/1194)	n/a[b]
GA	13 (566/570)[c]	17[d]	16 (809/810)[c]	> 17[d]

[a] All q-histograms are separated for every nonzero algebraic γ (Theorem 2(ii)). [b] R is not on Π. [c] GA lies on Π and merges no q-histograms below order 17, so there it ties the pure-γ points. [d] The witness (Section 3.1) has maximum degree 5; among trees with $\Delta \leq 4$, GA merges no q-histograms up to order 17, the largest examined.

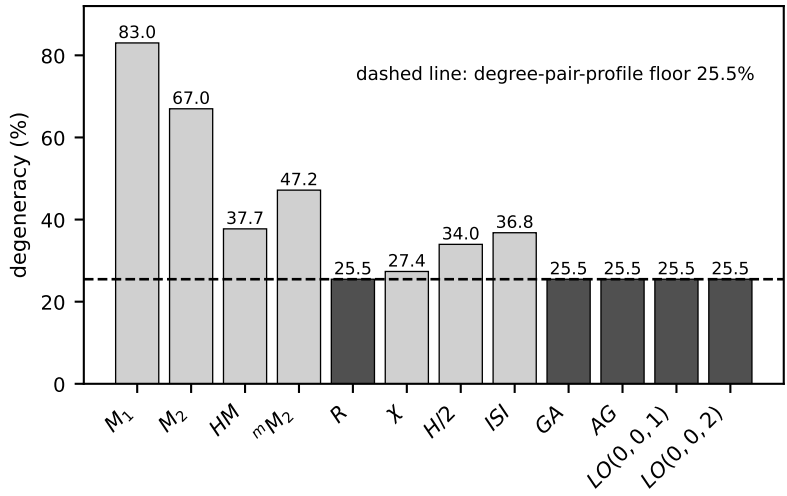


Figure 1. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10. The dashed line marks the degree-pair-profile floor, $25.5\% = 100(1 - 79/106)$. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so no vertex-degree-based index can fall below it; darker bars mark the indices on the floor. An index on the floor separates every pair of trees with distinct profiles at this order; it carries no information beyond the profile, and the pure-γ points share the floor with R, GA and AG.

their imbalances separate them. On all trees of order 16 (4069 profiles), $LO(0, \frac{1}{2}, 0)$ and $LO(0, \frac{1}{3}, 0)$ take only 3475 and 3476 distinct values, while $LO(0, \frac{1}{2}, 1)$ and $LO(0, \frac{1}{3}, 1)$ separate all 4069 profiles. $LO(\frac{1}{2}, 0, \cdot)$ rises from 3940 to 4067 and the Randić point $LO(-\frac{1}{2}, 0, \cdot)$ from 4007 to 4068 as γ goes from 0 to 1. Points on Π such as GA are unaffected (3996 in both cases), consistent with Theorem 2, and generic parent points such as $LO(0.3, 0.2, 0)$ already separate all 4069 profiles (Proposition 8). These counts are exact for orders 12–16 (at $\gamma = 1$ by Proposition 3 and [7]), but the effect is not proved for larger orders. It yields no predictive gain over a tuned parent in Section 6, and at fixed fractional exponents the sign of the change depends on the property (Supplementary Section S5).

4.3 Structure sensitivity

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [16], with a fingerprint-based variant due to Rakić and Furtula [42], was implemented with the published protocol on the 75 decane isomer trees. For a tree, the similar structures are those obtained by relocating a single edge within the molecular class. SS and Abr are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [6], with eleven of fourteen SS/Abr values agreeing exactly at the four displayed decimals. Each of the other three agrees within one unit in the fourth. The protocol and this control are detailed in Supplementary Section S6. Under the published desideratum (SS large, Abr small), $LO(0,0,2)$ has a structure sensitivity indistinguishable at three decimals from that of the hyper-Zagreb index (0.1257 against 0.1256). Its abruptness is markedly smaller (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure-γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451, Table 3). In view of the redundancy noted in Section 6.1, these third-decimal differences do not rank the indices. SS is also not a fixed property of the family. Along the pure-γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$), while the ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). The pure-γ points reduce to first order in γ to the parameter-free aggregate $IRLA$, since $LO(0,0,\gamma) = m + \frac{\gamma}{2}\, IRLA + O(\gamma^2)$, and this aggregate has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). This comparable ratio of the first-order aggregate suggests that the sensitivity–abruptness balance does not depend on the exponential form. The equality of SS with hyper-Zagreb at $\gamma = 2$ reflects the chosen value of γ.

5 Closed forms, bounds and extremal trees

5.1 Closed-form values on standard graphs

The classical graph families used are the complete graph K_n, the cycle C_n, the path P_n, the star S_n and the complete bipartite graph $K_{a,b}$. Beyond

Table 3. Structure sensitivity SS, abruptness Abr, and their ratio on the 75 decane isomers under the published similar-structure protocol of [16] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1256	0.3126	0.4017
${}^{m}M_2$	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0535	0.4507
AG	0.0256	0.0579	0.4415
$LO(0,0,1)$	0.0624	0.1367	0.4565
$LO(0,0,2)$	0.1257	0.2745	0.4579

these, the hypercube Q_k has 2^k vertices and $k\cdot 2^{k-1}$ edges and is k-regular. The wheel W_n is an $(n-1)$-cycle plus a universal vertex, and the friendship graph F_p consists of p triangles sharing a common vertex. We also treat r-regular graphs.

Proposition 9 (Closed forms)**.** *Let $\alpha,\beta,\gamma\in\mathbb{R}$. Then:*

1. $LO(K_n;\alpha,\beta,\gamma)=\dfrac{n(n-1)}{2}\,2^{\beta}(n-1)^{2\alpha+\beta}$ *for $n\geq 2$.*
2. $LO(C_n;\alpha,\beta,\gamma)=n\cdot 4^{\alpha+\beta}$ *for $n\geq 3$.*
3. $LO(P_n;\alpha,\beta,\gamma)=\begin{cases}2^{\beta}, & n=2,\\ 2\cdot 2^{\alpha}3^{\beta}e^{\gamma/3}, & n=3,\\ 2\cdot 2^{\alpha}3^{\beta}e^{\gamma/3}+(n-3)\cdot 4^{\alpha+\beta}, & n\geq 4.\end{cases}$
4. $LO(S_n;\alpha,\beta,\gamma)=(n-1)\,(n-1)^{\alpha}n^{\beta}\exp\!\left(\gamma\,\dfrac{n-2}{n}\right)$ *for $n\geq 2$.*
5. $LO(K_{a,b};\alpha,\beta,\gamma)=ab\,(ab)^{\alpha}(a+b)^{\beta}\exp\!\left(\gamma\,\dfrac{|a-b|}{a+b}\right)$ *for $a,b\geq 1$.*
6. $LO(Q_k;\alpha,\beta,\gamma)=k\cdot 2^{k-1}\cdot 2^{\beta}k^{2\alpha+\beta}$ *for $k\geq 1$.*
7. *For $n\geq 5$, W_n has $n-1$ spoke edges with degree pair $(3,n-1)$ and*

$n-1$ *rim edges with degree pair* $(3,3)$*, so*

$$LO(W_n;\alpha,\beta,\gamma) = (n-1)\Big[\,3^{\alpha}(n-1)^{\alpha}(n+2)^{\beta}\exp\Big(\gamma\,\tfrac{n-4}{n+2}\Big) + 9^{\alpha}6^{\beta}\Big].$$

8. *For* $p \geq 1$*,* F_p *has* $2p$ *spoke edges* $(2,2p)$ *and* p *outer edges* $(2,2)$*, so*

$$LO(F_p;\alpha,\beta,\gamma) = p\cdot 4^{\alpha+\beta} + 2p\,(4p)^{\alpha}(2p+2)^{\beta}\exp\Big(\gamma\,\tfrac{p-1}{p+1}\Big).$$

9. *For an* r*-regular graph* G *with* $r \geq 1$ *and* m *edges,* $LO(G;\alpha,\beta,\gamma) = m\cdot 2^{\beta}r^{2\alpha+\beta}$*.*

Proof. Each case follows by enumerating the edge degree pairs and applying Definition 1. In K_n, C_n, Q_k and r-regular graphs every edge has $d_u = d_v = r$, so imb $= 0$ and $(d_u d_v)^{\alpha}(d_u+d_v)^{\beta} = 2^{\beta}r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n-1$ edges of S_n have degree pair $(1,n-1)$ with imb $= (n-2)/n$ (item (4)), and all ab edges of $K_{a,b}$ have pair (a,b) (item (5)). For $n \geq 4$, P_n has two pendant edges $(1,2)$ and $n-3$ interior edges $(2,2)$, and the cases $n = 2,3$ are immediate (item (3)). Items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p. ∎

5.2 Extremal trees for the pure-imbalance points

Proposition 10 (Extremal trees of the pure-γ points)**.** *Let* $n \geq 4$ *and* $\gamma > 0$*. Among all trees of order* n*, the path* P_n *is the unique minimiser and the star* S_n *the unique maximiser of* $LO(\cdot\,;0,0,\gamma)$*, with*

$$(n-3) + 2e^{\gamma/3} \;\leq\; LO(T;0,0,\gamma) \;\leq\; (n-1)\,e^{\gamma(n-2)/n}.$$

For $\gamma < 0$ *the roles of* P_n *and* S_n *are exchanged.*

Proof. Let $\gamma > 0$. Every edge uv of a tree of order n has $q_{uv} \leq (n-2)/n$, with equality if and only if $\{d_u, d_v\} = \{1, n-1\}$, so $LO(T;0,0,\gamma) \leq (n-1)e^{\gamma(n-2)/n}$ with equality if and only if every edge is a $(1,n-1)$-edge, that is, $T = S_n$. For the lower bound, a tree with $k \geq 2$ leaves has at least k edges incident to a leaf, and each such edge has $q \geq \frac{1}{3}$ because

the other end has degree at least 2 and $(d-1)/(d+1)$ is increasing in d. Hence $LO(T;0,0,\gamma) \geq (n-1-k) + ke^{\gamma/3} \geq (n-3) + 2e^{\gamma/3}$, since $e^{\gamma/3} > 1$. Equality in the second inequality forces $k = 2$, so $T = P_n$, and P_n attains the bound because its two pendant edges are $(1,2)$-edges and its other edges are $(2,2)$-edges. For $\gamma < 0$ every inequality reverses. ■

Proposition 10 is a consequence of the criteria of Cruz and Rada [12]; the direct proof above is included because it is short. For the weights $\varphi_{ij} = e^{\gamma q_{ij}}$ and $E = e^{\gamma/3}$, their function $f(i,j) = \frac{ij}{i+j}(\varphi_{ij} + 2\varphi_{12} - 3\varphi_{22})$ equals $2(E-1)$ at $(1,2)$ and $(2,2)$. For $\gamma > 0$ it is larger at every other pair: if $i,j \geq 2$ and $(i,j) \neq (2,2)$ then $\varphi_{ij} + 2E - 3 \geq 2(E-1) > 0$ and $ij/(i+j) > 1$, and for $(1,j)$, $j \geq 3$, one has $f \geq \frac{3}{4}(e^{\gamma/2} + 2E - 3) > 2(E-1)$, since with $x = e^{\gamma/6} > 1$ this reads $3x^3 - 2x^2 - 1 = (x-1)(3x^2 + x + 1) > 0$. Theorem 2.1 and Corollary 2.2 of [12] then give the lower bound with equality only for P_n, and their Theorem 3.1 and Corollary 3.2 give the upper bound with equality only for S_n, because $e^{\gamma q}$ is largest at the pair $(1, n-1)$. For $\gamma < 0$ every inequality reverses. Gao and Gao [17] give general sufficient conditions for the path to be the unique minimal tree and the star the unique maximal tree of an exponential vertex-degree-based index, a class to which $LO(\cdot\,;0,0,\gamma) = \sum_{uv} e^{\gamma q_{uv}}$ belongs. The star half also follows from the sufficient condition of [40, Section 4.2]; the path half does not follow from the corresponding condition there, because the $(1,2)$-weight $e^{\gamma/3}$ exceeds the $(2,3)$-weight $e^{\gamma/5}$. The proof uses only the bounds on q and does not extend verbatim to general (α, β), where the ordering of trees depends on the scale exponent $2\alpha+\beta$ as well as on γ. Among chemical trees ($\Delta \leq 4$) the termwise bound gives $LO(T;0,0,\gamma) \leq (n-1)e^{3\gamma/5}$ for $\gamma > 0$, with equality only for $n = 5$. The path remains the unique minimiser for $\gamma > 0$ among chemical trees by Proposition 10, while the maximal chemical trees for $n > 5$ are left open. The orders 4 to 12 were also checked by enumeration in the reproducibility package.

5.3 Termwise bounds

Throughout this subsection G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. For $\gamma \geq 0$ every edge satisfies $1 \leq e^{\gamma q_{uv}} \leq$

$e^{\gamma(\Delta-1)/(\Delta+1)}$. At $\gamma = 0$ bounds of the types below, with their equality cases, are known [18, 33, 38], and inequalities between exponential indices are given in [10]. The statements below extend them to $\gamma \neq 0$ (Propositions 12 and 13 to $\gamma > 0$ only) and track how γ changes the equality cases. The inequalities of the Cauchy–Schwarz interpolation corollary (Section 5.3) and of Propositions 12 and 13, not their equality cases, are machine-checked. For Propositions 11 and 14 only the summation step is, with the edgewise extrema taken as hypotheses (Supplementary Section S7).

Definition 4 (Edge weight)**.** For real parameters α, β, γ and positive reals x, y,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^{\alpha}(x + y)^{\beta} \exp\Big(\gamma \, \tfrac{|x-y|}{x+y}\Big). \tag{5}$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 11 (Bracket bound)**.** *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \;\leq\; LO(G; \alpha, \beta, \gamma) \;\leq\; m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \tag{6}$$

where $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$. The lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$, hence $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$ on every edge, and summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (6). Each side of (6) is the sum of m per-edge inequalities, and is therefore an equality if and only if every per-edge inequality is. That is, equality holds if and only if every edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta, \Delta)$. ∎

Corollary (Cauchy–Schwarz interpolation)**.** *Let $\alpha_1+\alpha_2=2\alpha$, $\beta_1+\beta_2=2\beta$, and $\gamma_1+\gamma_2=2\gamma$. Then for every simple connected graph G with $m\geq 1$ edges,*

$$LO(G;\alpha,\beta,\gamma)\ \leq\ \sqrt{LO(G;\alpha_1,\beta_1,\gamma_1)\cdot LO(G;\alpha_2,\beta_2,\gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u,d_v;\alpha_1,\beta_1,\gamma_1)/\psi(d_u,d_v;\alpha_2,\beta_2,\gamma_2)$$

is the same for every edge of G.

Proof. For each edge $e=uv$ write $\psi_e^{(k)}:=\psi(d_u,d_v;\alpha_k,\beta_k,\gamma_k)>0$, $k\in\{1,2\}$, and by additivity of the exponents and the hypotheses on $(\alpha_k,\beta_k,\gamma_k)$, $\psi_e^{(1)}\psi_e^{(2)}=\psi(d_u,d_v;\alpha,\beta,\gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e=(\psi_e^{(1)})^{1/2}$ and $b_e=(\psi_e^{(2)})^{1/2}$ gives $\sum_e a_eb_e\leq(\sum_e a_e^2)^{1/2}(\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_eb_e=LO(G;\alpha,\beta,\gamma)$, $\sum_e a_e^2=LO(G;\alpha_1,\beta_1,\gamma_1)$ and $\sum_e b_e^2=LO(G;\alpha_2,\beta_2,\gamma_2)$. Equality holds if and only if $b_e=\lambda a_e$ for a constant $\lambda\geq 0$ and all e, i.e. if and only if $\psi_e^{(2)}/\psi_e^{(1)}=\lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta\mapsto LO(G;\theta)$, Theorem 1.) ∎

Proposition 12 (Upper bound in m and Δ)**.** *Let G be a connected graph of size $m\geq 1$ and maximum degree Δ. For $\alpha,\beta\geq 0$ and $\gamma\geq 0$,*

$$LO(G;\alpha,\beta,\gamma)\ \leq\ m\cdot\Delta^{2\alpha}\cdot(2\Delta)^{\beta}\cdot\exp\!\left(\gamma\,\frac{\Delta-1}{\Delta+1}\right). \tag{7}$$

For $\Delta=1$ the graph is K_2 and the bound is attained for every (α,β,γ). For $\Delta\geq 2$, equality holds if and only if $\alpha=\beta=\gamma=0$, or $\gamma=0$, $\alpha+\beta>0$ and G is Δ-regular, or $\gamma>0$, $\alpha=\beta=0$ and $G\cong K_{1,\Delta}$; for $\Delta\geq 2$, $\gamma>0$ and $\alpha+\beta>0$ the inequality is strict.

Proof. On every edge uv we have $1\leq d_u,d_v\leq\Delta$, hence $d_ud_v\leq\Delta^2$ and $d_u+d_v\leq 2\Delta$, and the normalised imbalance $\mathrm{imb}(x,y)=|x-y|/(x+y)$ is maximised over $\{(i,j):1\leq i\leq j\leq\Delta\}$ at $(1,\Delta)$, so $|d_u-d_v|/(d_u+d_v)\leq$

$(\Delta-1)/(\Delta+1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^\alpha$, $t \mapsto t^\beta$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha}(2\Delta)^\beta \exp(\gamma(\Delta-1)/(\Delta+1))$, and summing over the m edges yields (7). For $\gamma = 0$ the exponential factor is 1. If $\alpha = \beta = 0$ both sides equal m, and if $\alpha + \beta > 0$ equality forces $d_u d_v = \Delta^2$ or $d_u + d_v = 2\Delta$ on every edge, that is $d_u = d_v = \Delta$, so G is Δ-regular; the converse is immediate. For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G, $G \cong K_{1,\Delta}$). If $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

Proposition 13 (Lower bound in m and δ)**.** *Let G be a connected graph of size $m \geq 1$ and minimum degree δ. For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \;\geq\; m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \tag{8}$$

with equality if and only if G is δ-regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$ and $(d_u + d_v)^\beta \geq (2\delta)^\beta$. For $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative, and multiplying and summing over the m edges yields (8). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the bound for every graph, while otherwise equality requires equality in every active per-edge factor, which for $\alpha > 0$ or $\beta > 0$ forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ-regular. Conversely, on a δ-regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below. The bound is in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition 14 (Ratio bound)**.** *Let G be a simple graph with $E(G) \neq \varnothing$ and maximum degree at most Δ. Let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a*

BID index with edge contribution $h(i,j)$ strictly positive on every degree pair (i,j) that appears in G. Define

$$c^h_{\min} = \min_{(i,j)} \frac{\psi(i,j;\alpha,\beta,\gamma)}{h(i,j)}, \qquad c^h_{\max} = \max_{(i,j)} \frac{\psi(i,j;\alpha,\beta,\gamma)}{h(i,j)},$$

where the extrema range over the degree pairs actually appearing in G. Then

$$c^h_{\min} \cdot I_h(G) \;\le\; LO(G;\alpha,\beta,\gamma) \;\le\; c^h_{\max} \cdot I_h(G), \tag{9}$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h.

Proof. On every edge uv we have $h(d_u,d_v) > 0$ and $c^h_{\min} \le \psi(d_u,d_v;\alpha,\beta,\gamma)/h(d_u,d_v) \le c^h_{\max}$. Multiplying by $h(d_u,d_v)$ and summing over $E(G)$, with $\sum_{uv} h(d_u,d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G;\alpha,\beta,\gamma)$, gives (9). Each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 14 is a *graph-specific* comparison, whose constants depend on the degree pairs that actually appear in G. Its constants are the best edgewise ones, but a bound is attained only when every edge attains the extremum. A uniform version follows by enlarging the range of the extrema.

Corollary (Uniform ratio bound)**.** *Assume the hypotheses of Proposition 14, but with h strictly positive on all of $P_\Delta = \{(i,j) : 1 \le i \le j \le \Delta\}$ and the extrema taken over P_Δ. Then inequality* (9) *holds simultaneously for every graph of maximum degree at most Δ, with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ. The uniform constants are never better than the graph-specific ones, and are strictly worse unless G contains a degree pair attaining the corresponding extremum over P_Δ.*

6 QSPR diagnostics

We compare the family with the classical indices it reduces to on the 18 octane isomers, following the benchmark of the diminished Sombor study [34], and use the 35 nonane isomers as a replication set. The comparison set consists of the exact reductions $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$ and the fixed pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$. The edge count $m = LO(0,0,0)$ is constant on the octanes and is omitted. No parameter in the correlation comparison is tuned against the property data. The repository history begins with the octane correlations already computed, so it cannot document that the benchmark points $\gamma = 1$ and 2 were fixed before the octane data were seen. An exploratory local-tuning check (Supplementary Section S2) exceeded the best fixed point only by a 0.001 in-sample ΔH_{vap} margin and, under nested leave-one-out, on octane T_B (0.764 against 0.745), a small point-estimate difference without an interval; no superiority is claimed. The sets are small, so the octane correlations are descriptive, and the pre-registered tests on measured boiling points of Section 6.4 are the primary predictive evidence.

6.1 Correlations on the octane isomers

Table 4 reports the Pearson correlation r between each index and the boiling point T_B, the enthalpy of formation ΔH_f, the enthalpy of vaporisation ΔH_{vap}, the entropy S and the acentric factor ω of the 18 octane isomers. The property values are the standard benchmark compilation, audited against the NIST WebBook [31] and the 1947 API Research Project 44 tables [46] (Supplementary Table S5 and Section S1). The open items are seven entropies, the compiled acentric factors and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane, which is crystalline at 298 K. The benchmark value 8.41 kcal mol^{-1} has no stated source and lies just below the estimate for a hypothetical supercooled liquid from sublimation minus fusion enthalpy (8.56). The NIST value of 10.26 at 298 K describes sublimation (Supplementary Section S8). In the ΔH_{vap} column $LO(0,0,1)$ stays first, level at three decimals with $LO(0,0,2)$, with the

isomer excluded ($|r| = 0.980$) and with the hypothetical-liquid estimate ($|r| = 0.976$), while the sublimation enthalpy lowers every $|r|$ to 0.374 or below. These correlations are therefore conditional on a liquid-phase value for this isomer. The entropies can move the 0.001 margin between HM and M_1, so the ordering of the S column is not interpreted. None of the open items enters the tests of Sections 6.3 and 6.4.

Table 4. Pearson correlation r between the Loyola reductions, the fixed pure-γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest $|r|$ per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (Supplementary Section S1). These intervals are post-selection and not adjusted for multiplicity. The entry −0.000 (ISI, ΔH_f) is −0.0002.

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0,1,0)$	−0.718	−0.762	−0.935	−0.970	−0.970
$M_2 = LO(1,0,0)$	−0.497	−0.542	−0.811	−0.934	**−0.988**
$HM = LO(0,2,0)$	−0.654	−0.710	−0.903	**−0.971**	−0.982
${}^{m}M_2 = LO(-1,0,0)$	**+0.855**	**+0.891**	+0.925	+0.863	+0.794
$R = LO(-1/2,0,0)$	+0.819	+0.850	+0.958	+0.932	+0.898
$\chi = LO(0,-1/2,0)$	+0.800	+0.832	+0.961	+0.947	+0.924
$H/2 = LO(0,-1,0)$	+0.821	+0.849	+0.961	+0.932	+0.900
$ISI = LO(1,-1,0)$	−0.005	−0.000	−0.383	−0.593	−0.738
$GA = 2\,LO(1/2,-1,0)$	+0.822	+0.858	+0.965	+0.939	+0.906
$AG = \frac{1}{2}LO(-1/2,1,0)$	−0.816	−0.859	−0.957	−0.938	−0.900
$LO(0,0,1)$	−0.828	−0.832	**−0.984**	−0.928	−0.929
$LO(0,0,2)$	−0.829	−0.846	−0.981	−0.937	−0.925

The best correlation is split. ${}^{m}M_2 = LO(-1,0,0)$ is strongest on T_B and ΔH_f, $HM = LO(0,2,0)$ on S, $M_2 = LO(1,0,0)$ on ω and the fixed point $LO(0,0,1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0,0,2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the classical indices, as they must. Deleting single isomers shifts each bolded $|r|$ by at most 0.027 (Supplementary Section S1). The descriptors are nearly collinear. Fourteen pairs have $|r| \geq 0.99$, the first two principal components carry 99.05% of the variance (Supplementary Section S1), and the largest $|r|$ with a reduction is 0.982 for $LO(0,0,1)$ and 0.993 for $LO(0,0,2)$. Differences in the third decimal are therefore not rankings, and the bolded entries, each the largest of twelve correlations on 18 molecules,

are optimistic.

Proposition 7 is realised on these data. The 18 isomers have only 16 distinct count vectors, and the members of the two collision pairs of Table 1 differ by 1.2 °C and 2.1 °C in T_B although every vertex-degree-based index gives them equal values.

6.2 Effect of freeing γ and of the search box

We ask whether, at identical search budgets, freeing γ improves out-of-sample prediction over an optimised parent $M_{\alpha,\beta}$. The design is a fully nested leave-one-out (Supplementary Section S3). For each seed, candidate vectors are drawn uniformly from a search box, triples (α, β, γ) for the Loyola model and pairs (α, β) with $\gamma = 0$ for the parent, at equal budget. They are scored on the training isomers by the inner leave-one-out root-mean-square error of a one-descriptor linear regression, and the best is refit and used to predict the held-out isomer.

In the box $(-2, 2)$ freeing γ improved the nested Q^2 for ΔH_{vap} in 100/100 candidate streams at budgets 200 and 500, with median $\Delta Q^2 = +0.069$ and $+0.068$ (Supplementary Table S3), and in 93/100 and 96/100 with 2,2,3,3-tetramethylbutane excluded. It usually degraded ΔH_f and was mixed elsewhere. For ΔH_{vap}, however, the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds. By $\ln(ij) = 2\ln(i+j) + \ln(1-q^2) - \ln 4$ (Eq. (4)) the parent reaches imbalance information through $(1-q^2)^{\alpha}$ along $\beta \approx -2\alpha$, which needs large exponents and is the direction the box truncates. On a dense parent grid (steps 0.05, 0.1, 0.2 and 0.25) the nested Q^2 of the parent for octane ΔH_{vap} is 0.8873, 0.9156, 0.9525 and 0.9383 at half-widths 2, 4, 8 and 12. The selected point lies within one grid step of the boundary in 18, 18, 18 and 0 of the 18 folds. On the nonanes the values are 0.8532, 0.8779, 0.8997 and 0.8837, with 34, 34, 34 and 0 boundary folds of 34. The half-width-12 box, chosen after inspecting these curves as the smallest examined in which the parent's selections are interior, is used below for every property. This box does not favour the parent uniformly. For T_B it lowers the parent's nested Q^2 from 0.665 to 0.606 on the octanes and from 0.576 to 0.524 on the nonanes.

The matched comparison was repeated, without pre-registration, with

2000 candidates per model, seeds 0–49 and boxes of half-width 2 and 12 (Supplementary Table S6). There Edge is the share of outer folds, pooled over the 50 seeds, in which the parent's selection lies within 10% of the boundary. In the wide box the vaporisation-enthalpy gain shrinks to about zero. A hybrid control widens only (α, β), keeps $\gamma \in (-2, 2)$ and is drawn after the parent from the same candidate stream. It gives median $\Delta Q^2 = +0.004$ on the octanes (28 of 50 seeds) and -0.002 on the nonanes (24 of 50), while widening all three coordinates gives -0.010 and -0.016. Most of the change comes from the parent improving in the wide box ($0.887 \to 0.942$ on the octanes), the rest from the three-dimensional search being slightly less efficient there ($0.957 \to 0.946$ for the hybrid control). On the other properties no advantage reaches the seed-count criterion in either box, and octane S moves from 11 to 38 wins, which shows how strongly the win count depends on the search protocol. The fixed point $LO(0, 0, 1)$, singled out on these octanes, and the parameter-free $IRLA$ match the fixed parent point $M_{4.5,-9}$, which lies near the parent's wide-box selections (leave-one-out $Q^2 = 0.957$ and 0.948 against 0.949, on the octanes). That point was read off those selections, so its Q^2 is post-selection; the nested wide-box parent gives 0.938 to 0.942. Under these protocols tuning γ gives no demonstrated advantage. Equal budgets do not give equally thorough searches in two and three dimensions, so the design may cost the three-parameter model part of the effect sought.

6.3 Replication on the nonane isomers

Under a protocol committed to the reproducibility package (commit `964b127`) before the nonane data were committed and before any model was fitted to them, the nested design was applied to the 35 constitutional nonane isomers. Boiling points were 35 values from the NIST WebBook (NIST averages where given, otherwise the most recent determination, a deviation recorded in the protocol). Standard vaporisation enthalpies were 34 values from the NIST WebBook [31], of which 29 are compiled values: NIST lists 4 under Labbauf et al. [29] (printed as Labauf) and 25 under Reid [44], a review of the handbook [54]. Agreement with them therefore measures consistency with compiled values (Supplementary

Section S4). The pre-registered verdict is that the γ effect *replicates* for ΔH_{vap}, with LO better than GM in 95/100 and 98/100 seeds and median $\Delta Q^2 = +0.048$ and $+0.053$, and is *mixed* for T_B (58/100 and 63/100). *Replicates* is the protocol's name for its seed-count rule, which the protocol itself describes as a summary of optimiser variance and not an inferential test; the molecule-bootstrap interval reported at the end of this subsection includes 0. Within-fold baselines are in Supplementary Table S7. The protocol fixed the box of half-width 2, which binds the parent, and in the wide box the gain shrinks to about zero (median $\Delta Q^2 = -0.002$ under the hybrid control). We report the verdict as obtained. Its seed-count criterion counts optimiser variance on fixed data, which we regard as a design error, and after this result we made the measured boiling points of Section 6.4 the primary predictive evidence.

On both sets the untuned point $LO(0, 0, 1)$ matches or beats the tuned Loyola model in point estimate (0.957 and 0.903 against medians 0.953 and 0.900); on the octanes this point was singled out from the same data (Table 4). Since $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2}\, IRLA(G) + O(\gamma^2)$ as $\gamma \to 0$, it is closely related to $IRLA$; the correlation between the two is 0.996 on both sets, and $IRLA$ comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). A term linear in the imbalance suffices for this property in point estimate, but it is not necessary: the wide-box parent, which has no such term, comes close through $(1 - q^2)^{\alpha}$ (Section 6.2). Ridge regression on the degree-pair counts, a reference for what the counts carry, has a higher point estimate than every index model on the nonanes ($Q^2 = 0.927$ for ΔH_{vap}, 0.682 for T_B), although the conditional interval for LO minus ridge on nonane ΔH_{vap}, $[-0.101, +0.032]$, includes zero. On the octanes ridge is below the best index model for four of the five properties and level on ΔH_{vap} (0.954 against 0.957 for $LO(0, 0, 1)$; Supplementary Table S7). A post hoc molecule bootstrap conditional on the fitted predictions gives a 95% interval for LO minus GM on nonane ΔH_{vap} of $[-0.006, +0.113]$ at budget 200 (Supplementary Section S4).

6.4 Measured boiling points

Measured vaporisation enthalpies are too scarce for a larger test. Among the acyclic alkanes C4–C16, NIST lists only 25 skeletons outside C_8 and C_9, among them one of the 75 decanes, with at least one calorimetric standard-state 298 K determination. Two tests therefore used measured boiling points from the NIST WebBook [31], extracted by a rule committed to the repository (commit `a5cf378`) before any model was fitted. The rule admits all hexanes, heptanes and octanes, 34 of the 35 nonanes and 34 of the 75 decanes. The included and excluded decanes do not differ detectably in branching or imbalance (Supplementary Section S9), but with 34 and 41 molecules this does not show that the decane cohort is representative. Test A refits and reselects inside each leave-one-out fold on the decanes. Test B pools all 100 molecules with the carbon number n_C as a covariate. The octane and nonane boiling points were used above, so Test B is a pooled size-adjusted analysis rather than a fresh external test. Both use the wide-box design of Section 6.2, with $(\alpha, \beta) \in (-12, 12)^2$ for the parent and the same box with $\gamma \in (-2, 2)$ for the Loyola model, 2000 candidates and seeds 0–49. The pre-registered criterion for a γ effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero (Table 5), which we read as requiring the lower end of the two-sided 95% interval to exceed zero.

Table 5. Measured boiling points (nested leave-one-out Q^2). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number n_C in every index model (Class., $IRLA$, LO_1: n_C plus that descriptor); Ridge uses the counts alone, whose sum is $n_C - 1$. Base: training-fold mean (Test A) or n_C alone (Test B). Class.: best classical index chosen in each fold. LO_1: fixed $LO(0, 0, 1)$. Ridge: ridge regression on the degree-pair counts. GM, LO_h: medians over seeds 0–49 in the wide boxes, LO_h being the hybrid control with (α, β) in the wide box and $\gamma \in (-2, 2)$. ΔQ^2: median over seeds of the paired difference LO_h minus GM. The registered intervals are conditional on the fitted predictions and omit refitting and selection uncertainty (see below). Last column: seeds (of 50) in which LO_h beats GM.

Test	Base	Class.	$IRLA$	LO_1	Ridge	GM	LO_h	ΔQ^2	>
decanes ($n = 34$)	−0.062	0.176	0.167	0.149	0.860	0.514	0.406	−0.072	15
C6–C10, with n_C ($n = 100$)	0.938	0.967	0.963	0.963	0.991	0.977	0.977	−0.0001	25

Both pre-registered verdicts are negative. On the decane cohort the median $\Delta Q^2 = -0.072$, LO is ahead in 15 of 50 seeds, and the 95% interval is $[-0.156, +0.001]$. In the pooled size-adjusted set the values are -0.0001, 25 of 50 and $[-0.003, +0.002]$. In the original box of half-width 2, reported as the pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465), and the Loyola model is marginally ahead on the pooled set (0.980 against 0.976). Seed by seed, that narrow-box Loyola model beats the wide-box parent GM in 38 of 50 seeds on the pooled set (median seed-matched difference $+0.0033$) but in only 22 of 50 on the decanes (median -0.037). Against the narrow-box parent the counts are 40 and 25 of 50. These comparisons were not registered, carry no interval and count optimiser variance on fixed data, so they do not indicate a gain from γ. The pooled cohort was corrected after an external audit (Table 6). On the uncorrected cohort of 101 molecules the same test gave -0.0001, 24 of 50, $[-0.003, +0.002]$, so the correction changes no conclusion. The intervals are conditional on the fitted predictions and omit refitting and selection. An exploratory resampling reruns the whole nested procedure for both models on random subsamples of 80% of the molecules. Each replicate uses a single candidate stream, so the ranges also contain optimiser variance. On the decanes (500 subsamples) the median ΔQ^2 is -0.110 with central 95% range $[-0.510, +0.312]$. On the pooled set (200 subsamples) it is $+0.001$ with range $[-0.015, +0.008]$. Neither range supports a gain from γ. The selection-aware decane range is so wide that it neither supports nor excludes gains of practical size, the registered interval omits refitting and selection, and no smallest effect of interest or equivalence margin was registered. The registered contrast also gives LO_h three dimensions to search against two for GM with the same candidate budget, which may handicap the three-parameter model (Section 6.2). The negative verdicts therefore mean that no gain was demonstrated under these protocols; they do not establish equivalence, and they do not show that γ never helps. The two summaries differ in training size and seed averaging and are not directly comparable. Nor are the ranges confidence intervals for the full-sample ΔQ^2. Ridge regression on the ten degree-pair counts reaches $Q^2 = 0.860$ on the decanes, where every single index

stays between 0.149 and 0.514 (medians over seeds for the tuned models), and 0.991 on the pooled set. On the registered secondary question of Test B, $IRLA$ and $LO(0,0,1)$ do not beat size plus the best classical index (0.963 against 0.967), while both tuned models have higher point estimates (0.977; no interval was registered for this secondary question). A permutation check on the decanes repeats the whole nested procedure, including the choice of the ridge penalty, for 200 random permutations of the boiling points. The permuted Q^2 has median -0.09, 95th percentile -0.01 and maximum 0.20. None reaches 0.860, so the empirical p-value is $1/201 \approx 0.005$. Wiener's two-descriptor model [53], with the path number W and Wiener's polarity number p_3, the number of carbon pairs three bonds apart, fitted without tuning, gives $Q^2 = 0.904$ on the decanes. On trees $p_3 = \sum_{uv}(d_u - 1)(d_v - 1)$, since a path of length three is fixed by its middle edge, so p_3 is a linear function of the degree-pair counts and the distance information in this model is W. This single point estimate, given without an interval and for descriptors historically chosen on alkane boiling points, suggests that within one carbon number W may add to the degree-pair information; its difference from ridge is not resolved here. Under the stated boxes and candidate budgets, ridge regression predicts the boiling points more accurately than every single-index model evaluated, with or without γ, in point estimate. The margin is large on the decanes and small on the pooled set, where size alone gives $Q^2 = 0.938$, and no interval for these differences is reported here. These conclusions concern acyclic alkanes.

7 Discussion and limitations

The results of Sections 3–5 are exact. Their limits, and those of Section 6, are as follows.

(i) The minimality of the collision orders in Theorem 2(iii), equality in Proposition 5(ii) for $\Delta \leq 6$, and the recovery of discrimination off Π in Section 4.2 rest on exhaustive enumeration. No general argument is given.

(ii) Whether two graphs with $\Delta \leq 4$ can exchange order five times along γ is open (see the remark on sharpness in Section 3.1).

(iii) The data sets are small and acyclic: 18 octanes, 35 nonanes, 34 decanes and 100 pooled C6–C10 alkanes. Seven octane entropies and the compiled acentric factors rest on secondary sources, and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane has an unresolved reference state (Supplementary Section S8).

(iv) Changes made after outcomes had been seen are listed in Table 6. None changes a pre-registered verdict as reported, but the wide-box analysis, in which the nonane gain shrinks to about zero, is itself not pre-registered.

Table 6. Changes to the pre-registered protocols made after outcomes had been seen. Protocols and change records are in the reproducibility package [35].

Test	Change after outcomes	Effect
Nonanes (Section 6.3)	Parent box widened from half-width 2 to 12, chosen after inspecting the ΔH_{vap} curves	Registered verdict reported as obtained; the gain shrinks to about zero in the wide box
Nonanes	Seed-count criterion judged to count optimiser variance	None on reported numbers
Nonanes	Measured boiling points made the primary predictive evidence	None on reported numbers
Boiling points, Test B	All NIST pages of a skeleton pooled; one nonane excluded, one heptane value moved by 0.2 °C	Verdict unchanged; both cohorts reported
Nonanes	A second NIST T_B of 3-ethyl-4-methylhexane, 6.4 K lower, found later	Still mixed if excluded or replaced (Suppl. S4)
Boiling points, Test B	Least-squares solver rescaled by columns	Verdict unchanged
Boiling points, Test B	Scale-invariant rank test	No displayed value changes
Boiling points, Test B	Stricter rank threshold and a deletion-rank guard in the size-adjusted scorer	No output changes
Boiling points, both	Interval rule read as one-sided	None
Boiling points, both	Selection-aware resampling added (Section 6.4)	Exploratory; no verdict depends on it

8 Conclusion

We extended the Gutman–Milovanović family by an exponential factor in the normalised degree imbalance of each edge. On the ratio plane the pure-

imbalance points at nonzero algebraic γ discriminate the histogram of edge imbalances and nothing finer, by Rada's theorem based on the Lindemann–Weierstrass theorem [39, Theorem 3.4], a limit that binds every index depending on an edge only through its degree ratio [39, Proposition 2.5]. By exhaustive enumeration, for trees this histogram carries the full degree-pair information up to order twelve and loses it at order thirteen, and at sixteen for chemical trees, with explicit colliding families. Along the imbalance axis two chemical graphs exchange order at most five times, the extremal trees of the pure-imbalance points are the path and the star, and at rational exponents a nonzero algebraic imbalance parameter never destroys a distinction the parent makes.

Under random search at matched budgets, no predictive gain from the imbalance parameter was demonstrated; this does not show that it never helps, and the decane intervals were wide. A gain on vaporisation enthalpies met the seed-count criterion of a pre-registered protocol, with a bootstrap interval that included zero. It shrank to about zero in an exploratory comparison once the parent was searched over a box that did not truncate it, and pre-registered tests on measured boiling points were negative. There a linear model on the degree-pair counts had higher point estimates than every single index. The untuned pure-imbalance point, close to the irregularity index $IRLA$, was as accurate as the tuned models on vaporisation enthalpy in point estimate, out of sample only on the nonanes, and no equivalence test was made. Three problems remain open: sharp two-sided bounds in terms of indices outside the family, such as the atom-bond-connectivity and Sombor indices; the extremal graphs for general parameter triples, beginning with the maximal chemical trees of the pure-imbalance points; and the behaviour on other compound classes.

Data availability. The supplementary sections and tables cited here are in the file `LOYOLA_supplement.pdf` of the reproducibility package [35]. Data, scripts and verifiers that recompute every tabulated value are in the reproducibility package [35], where the nonane and boiling-point protocols were committed before the corresponding models were fitted. The NIST WebBook pages from which the experimental values were extracted are not redistributed; a script of the package downloads them and checks each

against the copy from which the data sets were built. Lean 4 with Mathlib checks the termwise bounds, not their equality cases, and takes the edge-wise extrema of the bracket and ratio bounds as hypotheses. It also checks the convexity of the log-sum-exp form of $\log LO$, the spanning statement of the independence lemma and the order-thirteen profile collision on the pure-γ line. The transcendence, Laguerre and enumeration arguments are not machine-checked. Graphs are encoded there by their degree-pair data (Supplementary Section S7), and the file `LEAN_VERIFICATION.md` of the repository maps each statement of this paper to its Lean theorem.

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